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20.IMPLEMENT FUTURE SALES PREDICTION USING A SUITABLE MACHINE LEARNING ALGORITHM.

1. Future Sales Prediction – Linear Regression

Dataset 1: Monthly Sales

Month	Advertising	Sales
1	10	120
2	12	135
3	15	150
4	18	165
5	20	180
6	22	195
7	25	210
8	28	225

Program

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("sales1.csv")
X = df[["Month", "Advertising"]]
y = df["Sales"]
model = LinearRegression()
model.fit(X, y)
sample = [[9, 30]]
```

```
prediction = model.predict(sample)
print("Predicted Future Sales:", round(prediction[0], 2))
```

Output

Predicted Future Sales: 240.00

2. Future Sales Prediction – Decision Tree Regression

Dataset 2: Product Sales

Price	Advertising	Discount	Sales
100	10	5	120
95	12	10	140
90	15	10	160
85	18	15	180
80	20	20	200
75	25	20	230
70	28	25	250
65	30	25	270

Program

```
import pandas as pd
from sklearn.tree import DecisionTreeRegressor
df = pd.read_csv("sales2.csv")
X = df[["Price", "Advertising", "Discount"]]
y = df["Sales"]
model = DecisionTreeRegressor(random_state=1)
model.fit(X, y)
sample = [[68, 29, 25]]

prediction = model.predict(sample)
```

```
print("Predicted Future Sales:", round(prediction[0], 2))
```

Output

Predicted Future Sales: 270.00

3. Future Sales Prediction – Random Forest Regression

Dataset 3: Store Sales

Customers	Advertising	Holiday	Sales
100	5	0	200
120	7	0	240
140	8	0	280
160	10	1	350
180	12	1	390
200	15	1	430
220	18	0	440
250	20	1	520

Program

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor
df = pd.read_csv("sales3.csv")
X = df[["Customers", "Advertising", "Holiday"]]
y = df["Sales"]
model = RandomForestRegressor(
    n_estimators=50,
    random_state=1
)

model.fit(X, y)
```

```
sample = [[230, 19, 1]]
prediction = model.predict(sample)
print("Predicted Future Sales:", round(prediction[0], 2))
```

Output

Predicted Future Sales: 480.60

4. Future Sales Prediction – Polynomial Regression

Dataset 4: Seasonal Sales

Month	Temperature	Sales
1	20	110
2	22	125
3	25	145
4	28	170
5	30	195
6	32	220
7	34	245
8	33	235

Program

```
import pandas as pd
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
df = pd.read_csv("sales4.csv")
X = df[["Month", "Temperature"]]
y = df["Sales"]
poly = PolynomialFeatures(degree=2)

X_poly = poly.fit_transform(X)
```

```
model = LinearRegression()
model.fit(X_poly, ysample = [[9, 31]])
sample_poly = poly.transform(sample)
prediction = model.predict(sample_poly)
print("Predicted Future Sales:", round(prediction[0], 2))
```

Output

Predicted Future Sales: 250.84

5. Future Sales Prediction – Support Vector Regression

Dataset 5: Online Sales

WebsiteVisits	Offers	SocialMediaAds	Sales
1000	2	5	150
1200	3	6	175
1500	3	8	210
1800	4	10	250
2000	5	12	290
2300	5	14	320
2500	6	16	350
2800	7	18	390

Program

```
import pandas as pd
from sklearn.svm import SVR
from sklearn.preprocessing import StandardScaler
df = pd.read_csv("sales5.csv")
X = df[["WebsiteVisits", "Offers", "SocialMediaAds"]]
y = df["Sales"]

scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
model = SVR(kernel="rbf", C=100, gamma="scale")
model.fit(X_scaled, y)
sample = [[3000, 7, 20]]
sample_scaled = scaler.transform(sample)
prediction = model.predict(sample_scaled)
print("Predicted Future Sales:", round(prediction[0], 2))
```

Output

Predicted Future Sales: 367.52